ITCS216 Fundamentals of Database Systems (1/2026)

Faculty of Information and Communication Technology

Project Phase I: Database Design

(Total: 15% of Grade)

Description:

In this project, students will act as *database designers*, mostly the first step in the database career opportunities. Their main responsibility is to design the real database according to the selected business process/application.

Members: 5 Members

Requirements and Deliverables

There are 2 milestones across Phase I; each come with the deliverables:

Milestone 1 (M1): Requirement Analysis and Business Rules

In M1, students are required to analyze the selected business domain and list out the related business rules.

1. Select a business domain from the list below. Selection is on a first-come, first-served basis. To be noted that each section will have specific business domains and each domain will be given only 3 main business processes. This meant that across section will not have the same domain.
   (Register on Monday, Aug 10, 2026 12:30PM – Noon)

Section 1:

Business Domains	Business Processes
Hospitality business: Lodging (Hotel and Resort)	Room Reservation, Check-in (both Reserved and Walk-in), Check-out and Payment
Hospitality business: Hotel Restaurant	Food Ordering, Payment & Receipt, Stock management
Healthcare business: Spa and Fitness Centre	Membership Registration, Monthly Invoice, Payment & Receipt
Healthcare business: Hospital	Patient Admission, Treatment & Care, Payment & IPD (Inpatient Pharmacy)
Healthcare business: Beauty Clinique	In-Person Arrival & Check-in, Medical Consultation & Care, Check-out & Payment

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Section 2:

Business Domains	Business Processes
Real Estate & Space: Residential Apartment & Condo	Lease Agreement, Utility Billing, Maintenance Requests
Real Estate & Space: Commercial Office Building	Space Leasing, Access Management, Visitor Management
Real Estate & Space: Exhibition Center & Event Venue	Venue Reservation, Vendor Management, Event Utility Hookup Services
Real Estate & Space: Co-Living Space & Serviced Residence	Room Reservation, Resident Services, Checkout Billing
Real Estate & Space: Smart Parking & Garage Management	Parking Permit Management, Vehicle Access Control, Billing & Payment

Section 3:

Business Domains	Business Processes
Food business: Chain / Franchise Restaurants	Brand Registration, Monthly Invoice, Payment & receipt
Food business: Retail & Wholesale Distribution	Warehouse & Inventory Management, Ordering and Payment, Pick & Pack/Delivery
Food business: Processed & Ready-to-Eat Food Manufacturing	Stock Controlling, Food Preparation & Cooking, Payment & Receipt
Food business: Supply Chain, Logistics & Cold Storage	Procurement, Inventory Control, Transportation & Delivery
Food business: Agro-Industrial & Protein Production	Raw Material Acquisition, Logistic & Transport, Sales & Payment

2. Describe your selected business domain (e.g., history, their product and services, their platform) and the scope of your business domain in about 100-150 words.
3. Based on the assigned business domain and predefined business processes, explain on the workflow and detailed activities of each business process.
    - Based on your analysis, identify the business rules that govern the workflow.
    - Students may consult domain experts or conduct additional research to improve their understanding.
    - Supporting examples may be included where appropriate.
    - Students may introduce additional supporting business processes if necessary; however, the assigned business processes must remain as the core scope of the project, and any additional processes must be directly related to them.

4. List at least 30 business rules related to each business process in Step 3. The business rules must follow the given templates (see **Appendix**).
   *** Clarification: there will be at least 30 rules in total covering all 3 business processes ***
   Example

Process:	Book Purchasing
Business rules:	• Customers can make many orders. *(Template#1, M-N)* • Orders can contain many books. *(Template#1, M-N)* • Orders are paid with at most one payment method. *(Template#1, 1-M)* • One customer can have many credit cards registered in their account *(Template#1, 1-M)* • An order must have an order number, date and time purchased, and payment status *(Template#2)*

5. List all referenced websites or examples that you have collected the data.

Deliverables: A short report consists of Step 2-5
Submission: Sunday, August 30, 2026, at 11:59 PM

Milestone 2 (M2): Conceptual design with Entity Relationship Diagram

In M2, students are required to design and draw an entity relationship diagram from the selected business domain using **Chen's** notations. The requirements of the diagram and the design are listed as followed:

- The total number of entities **must** be at least 10 entities and we recommend not to exceed 15 entities. Part of these entities must be a specialization hierarchy, i.e., one super type and at least 2 sub types. **The specialization hierarchy (EERD) must include**:
  - Define at least one common attribute for the supertype entity
  - Define at least 2 unique attributes for each subtype entity
  - Define a subtype discriminator attribute and values.
  - Define the disjointness constraint: "disjointed" or "overlapping"
  - Define completeness constraint: "Total" or "Partial" completeness
- Each entity **must** have at least 3 attributes, including the key attribute.
  - Determine if the entities and relationships weak or strong (regular)
  - Define if the attributes are single or multi-valued, derived, or composite if any
- There must be at least 1 weak entity and 1 multi-valued attribute.
- Each relationship **must** specify the connectivity, cardinality, and participation constraints. There must be **at least one Many-to-Many** relationship.

Deliverables: A Progress Report consists of M1 and M2
Submission: Sunday, September 27, 2026, at 11:59 PM

Acknowledgement Usage of AI for Project

To support your learning, you are not allowed to use AI tools during workshop sessions or while completing project assignments. Practicing without AI will strengthen your understanding and skills. However, Grammarly may be used for editing project reports.

Penalties for late submission

If you hand in the work late, your marks will be reduced as follow:

a. A deduction of 10% of the maximum mark available from the actual mark achieved by you will be imposed upon expiry of the deadline. For example, 10% of 100 marks is 10 marks. If the deadline is 5pm and you submit at 6pm on the same day, your mark of 68 would become a mark of 58 after the penalty.

b. A further deduction of 5% of the maximum mark available from the actual mark achieved by you will be imposed on each of the next subsequent days.

Appendix

Analyze the business process and create business by following the templates:

Template#1: **Format:** Noun + Verb + Noun (e.g., a student enrolls many classes)

Business Process Name: Enrollment		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A student can enroll many classes.	1:M
Backward/Passive	A class can be enrolled by many students.	1:M
Conclusion	Students can enroll many classes.	M:N

The Backward and Forward rules give you intermediate result, while the "Conclusion" version give you a final version of your business rule

Template#2: **Format**: Noun + Verb + Attributes (e.g., a student comprises of ID, Name, Email),

Business Process Name: Enrollment
Rule No. 2
A student comprises of ID, Name, Email

Rubric for M1 (35 points)

Criteria	5 - Excellent:	3 - Satisfactory:	1 - Poor:	0- N/A
Introduction about selected business domain				
Quality	Comprehensive understanding of the industry, organization, goal, and the scope of the selected business domain	General understanding of the industry, organization, and goal, and the scope of the selected business domain	Lack of understanding of the industry, organization, and goal, and the scope of the selected business domain	No business domain introduction
Number of words	In range (100-150 words)	Out of range less than 25% (75-190 words)	Out of range more than 20% (<75 or >190 words)	
Reference	The references are correctly cited and presented	1-2 references are presented but incomplete	The references are incomplete	
Business process				
Completeness	All 5 business processes are clearly described and complete	At least 3 business processes are clearly described and complete	Only 1 or 2 business processes are clearly described and complete	No business process is present
Support evidence and references	Provide comprehensive support (e.g., example of documents/screenshots) and references.	Provide basic references without any solid supports	Provide unclear reference or supports	
Business rules:				
Informative	All 30 business rules are clear, concise, and related to the business process	All 20 business rules are clear, concise, and related to the business process	Less than 20 business rules are clear, concise, and related to the business process	No business rules are present
Well-structure	All 30 business rules follow the given templates	At least 20 business rules follow the given templates	Less than 20 business rules follow the given templates	

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Rubric for M2 (75 points)

Criteria	5 - Excellent	3 - Satisfactory	1 - Poor	0 - N/A
Entity				
Number of Entities including a specialization hierarchy type	Conform to the requirement of 10-15 entities) including 1 supertypes and at least 2-3 subtypes	Conform to the requirement of 10-15 entities but missing supertype or subtypes	Not conformed to the requirement e.g., no specialization hierarchy or less than 6 entities provided	No entities in the diagram
Design Quality	All entities conform to the business rules	1-2 entities conform to the business rules	At least half of the entities do not conform to the business rules	
Notation *Chen*	All entities are in the correct notation (including weak and strong entities)	1-2 entities are not in the correct notation (including weak and strong entities)	At least half of the entities are not in the correct notation (including weak and strong entities)	
Attributes				
Number of Attributes per entity (except the subtype entity)	All entities conform to the requirement of at least 5 attributes including the key attribute (including a multi-valued attribute)	Some entities do not conform to the requirement of at least 5 attributes including the key attribute (including a multi-valued attribute)	At least half of the entities do not conform to the requirement, i.e., less than 5 attributes per an entity (including a multi-valued attribute)	No attributes in the diagram
Design Quality	All attributes conform to the business rules and belong to the entity	Some attributes do not conform to the business rules or belong to the entity	All attributes do not conform to the business or belong to the entity	
Specialization hierarchy and constraints (Disjointness and Completeness)				
Design Quality	All supertype and subtype entities are correctly designed and conform to the business rule.	Some of the subtype entities or a supertype entity are incorrectly designed and/or partially conform to the business rule.	All supertype and subtype are incorrectly designed or do not conform to the business rule.	No specialization hierarchy and constraints in the diagram
Common / Unique attributes	All common and unique attributes are correctly designed and conform to the business rule.	Some of common or unique attributes are incorrectly designed and/or do not conform to the business rule.	At least half of common or unique attributes are incorrectly designed and/or do not conform to the business rule.	
Subtype discriminator	All subtype discriminator and its values are correctly specified in the diagram	The subtype discriminator is correctly defined and at least half of its values are correctly specified in the diagram	Missing subtype discriminator or all at least half of its values	
Disjointness and Completeness	All disjointness and completeness constraint are correctly designed and conform to the business rule	Some disjointness and completeness constraint are incorrectly designed and/or do not conform to the business rule.	All disjointness and completeness constraint are incorrectly designed and/or do not conform to the business rule.	
Relationship and its constraints (Connectivity, Cardinality, and Participation)				
Relationship includes at least	At least one M-to-N relationships are	-	At least one M-to-N relationships are defined	No relationship and its

Criteria	5 - Excellent	3 - Satisfactory	1 - Poor	0 - N/A
one M-to-N relationship	defined and conform to the business rule		but do not conform to the business rule	constraints in the diagram
Relationship connects to the correct entities based on the revised business rules	All relationships conform to the business rules (i.e., connect to the correct set of entities)	1-2 relationships do not conform to the business rules (i.e., connect to the incorrect set of entities)	At least half of the relationship do not conform to the business rules	
Relationship – *Connectivity*	All relationships are specified the correct type of connectivity constraints (i.e., 1-1, 1-M, and M-N) based on the business rules	1-2 relationships are specified the incorrect type of connectivity constraints (i.e., 1-1, 1-M, and M-N) against the business rules	At least half of relationships are specified the incorrect type of connectivity constraints (i.e., 1-1, 1-M, and M-N) against the business rules	
Relationship – *Cardinality*	All relationships are correctly specified cardinality constraints (i.e., min and max) based on the business rules or domain	1-2 relationships are incorrectly specified cardinality constraints (i.e., min and max) against the business rules or domain	At least half of relationships are incorrectly specified cardinality constraints (i.e., min and max) against the business rules or domain	
Relationship – *Participation*	All relationships are correctly specified participation constraints (i.e., mandatory or optional) based on the business rules or domain	1-2 relationships are incorrectly specified participation constraints (i.e., mandatory or optional) based on the business rules or domain	At least half of relationships are incorrectly specified participation constraints (i.e., mandatory or optional) based on the business rules or domain	
Notation *Chen*	All relationships and constraints are represented in notation	Some relationships and constraints are represented in incorrect notation	All relationships and constraints are represented in incorrect notation	